

Wingspan — an options agent that mostly says no

Alpaca AI Trading Agents Hackathon 2026 · team Convexity · paper account PA371G5THNUO · github.com/Jhosshua/OptionsAgent (MIT) · optionsagent-production.up.railway.app · snapshot Tue Sep 01, 23:29 ET

AI logic

One DeepSeek call per trading day (deepseek-v4-pro, JSON mode, temperature 0) over a 13-name sub-\$50 watchlist with recent bars and context. The output is a list of proposals, each {underlying, strategy_type, direction, conviction, thesis}, validated against a schema; anything malformed is dropped, never repaired. The model cannot name a strike, a size or an exit, and cannot place an order. A failed or empty call is journaled as a *proposer_result* row so a dead model can never pass for a quiet market.

Risk gates (deterministic, in code; environment may only tighten)

Conviction floor 0.60 · strategy menu (defined-risk credit spreads) · one position per underlying, watchlist names only, no re-entry on an open name · six concurrent positions · sizing = conviction-scaled share of buying power, hard-capped at \$3,000 of defined risk per position (≤ 15 contracts of a \$2 spread) · contract picker: short strike 0.15–0.30 delta, 30–45 DTE, width $\leq \$2$ · liquidity floor: credit $\geq \$0.10$ and unwind-now cost $< 1.5\times$ credit, so the exit rule cannot stop the trade out on its own entry quotes (replay of the prior day's chains: 8 of 22 naive picks were already past the stop; after the floor, 5 of 26 admitted, 0 past) · fills confirmed and the unfilled remainder cancelled before anything is booked. Exits, no AI, every 20 minutes: 50% profit target; $2\times$ credit stop confirmed over two sweeps after 10:00 ET; forced close at 21 DTE; a close is recorded only on a confirmed unwind fill. Paper-only by construction; every broker or model error fails closed and pages Discord. 276 tests; the gate, cap, fill-confirmation and CLI paths each have a mutation check that breaks a test when the guard is removed.

Alpaca infrastructure

Orders, positions, account and clock go through the official Alpaca CLI (*alpaca ... -q*, JSON out), pinned to v0.0.14 with a verified checksum in the Docker image, and journaled per call (argv, exit code, latency, order id). A spread is one *order submit --order-class mleg* limit order at the net credit; a lost reply is reconciled by our client order id before it is called a failure. Option chains come from a read-only market-data sidecar; stock bars from a hosted Alpaca market-data relay. Runs on Railway as Linux cron (entry 10:15 ET, exit sweep every 20 min, equity scalper every minute) with state on a volume; the public dashboard is a read-only observer of the same journals and renders the rails from the same functions the bot runs.

Results and disclosure

Competition account PA371G5THNUO: equity \$100,034.43; filled orders in the window 6 (0 options, 6 shares); active gate *research_rules*, cap \$3,000.00, broker path *cli*. The harness predates the event (July 2026). Built inside the window: the equity intraday scalper and its 6-month study, the Railway deployment, the DeepSeek proposer, the dashboard rework, the CLI transport, and the gate/cap changes with their adversarial review. The account was created 2026-08-30 with \$100,000 and nothing else has traded on it.